Reviewer Pack — Minimal Rank of Geometric Loci vs ACC0 Circuit Lower Bounds …

Entry 027f18151318 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-23 04:44:04 UTC

Minimal Rank of Geometric Loci vs ACC0 Circuit Lower Bounds for Explicit Functions

Entry ID: `027f18151318` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-23 04:44:04 UTC

1. Conjecture

Field A (mathematical branch): Real Algebraic Geometry **Field B** (complexity object): Complexity Theory: ACC0 Circuit Lower Bounds for Explicit Functions

Statement:

> [‘For any explicit function f in P, the minimal rank of its geometric locus in real projective space is at least Ω(n log n), where n is the size of the input.’, ‘Moreover, there exists a constructive mapping from an instance of ACC0 circuit lower bounds for explicit functions to a geometric locus in real projective space such that the rank of the geometric locus corresponds to the lower bound.’, ‘This implies that any polynomial-size ACC0 circuit can be realized with a geometric locus of at most O(n) dimension.’]

Rationale (proposer’s reasoning):

> [‘Real algebraic geometry has previously been used to understand the structure of polynomials and their relations. By applying it to explicit functions, we may uncover structural properties that are not evident in standard complexity theory.’, ‘The proposed connection suggests a way to translate a complexity-theoretic problem into an algebraic geometric one, which could potentially lead to new algorithms or insights.’, ‘If this conjecture holds, it would imply a deeper understanding of the boundaries between different complexity classes.’]

Taxonomy category: `ACC_SIPSER` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `d63fb28f2a403e58`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all generated explicit functions f in P with size $n \leq 40$, the rank of the geometric locus in real projective space is at least $\Omega(n \log n)$ across at least 80% of the seeds (24 or more out of 30), and the mean rank does not exceed $O(n)$.

The conjecture is falsified if any seed produces a rank that is less than Ω(n log n) or if the mean rank exceeds O(n).

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `Minimal Rank Geometric Loci real algebraic geometry AND ACC0 Circuit Lower Bounds explicit functions` - `ACC0 Circuit Lower Bound to Geometric Locus mapping real projective space rank comparison` - `Geometric Locus dimension O(n) ACC0 circuit complexity theory real algebraic geometry`

Top relevant hits considered: - [http://arxiv.org/abs/2304.02372v2] Explicit smooth real algebraic functions which may have both compact and non-compact preimages on smooth real algebraic - [http://arxiv.org/abs/2304.07540v2] On real algebraic maps whose images are domains surrounded by the products of hyperbolas and real affine spaces - [http://arxiv.org/abs/math/0103170v1] Minimizing Polynomial Functions - [http://arxiv.org/abs/1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [http://arxiv.org/abs/0708.0766v2] Invariant rigid geometric structures and smooth projective factors - [http://arxiv.org/abs/1010.1920v3] Tight lower bound to the geometric measure of quantum discord - [http://arxiv.org/abs/2510.17487v1] Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observin - [http://arxiv.org/abs/1411.4413v2] Observation of the rare $B_s^0 \to \mu^+\mu^-$ decay from the combined analysis of CMS and LHCb data

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        max_row = i
        for j in range(i+1, n):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]
        pivot = A[i][i]
        for j in range(i, n + 1):
            A[i][j] /= pivot
```

```
        for k in range(n):
            if k != i:
                factor = A[k][i]
                for j in range(i, n + 1):
                    A[k][j] -= factor * A[i][j]
    return A

def rank(A):
    A = [row[:] for row in A]
    r = gaussian_elimination(A)
    rank = sum(1 for row in r if any(row[j] != 0 for j in range(len(row))))
    return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    f = [random.uniform(-1, 1) for _ in range(n)]

    # Construct the geometric locus
    A = [[f[j] ** i for j in range(n)] for i in range(n + 1)]
    rank_value = rank(A)

    metric_name = "rank"
    metric_value = rank_value
    instances_tested = 1
    conjecture_holds = rank_value >= n * math.log(n, 2)
    counterexample = "" if conjecture_holds else f"Rank {rank_value} < Ω({n} log {n})"

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fract
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9239fd2e.py", line 71, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9239fd2e.py", line 49, in run_trial
    rank_value = rank(A)
                 ^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9239fd2e.py", line 38, in rank
    r = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9239fd2e.py", line 28, in gaussian_elim
    A[i][j] /= pivot
    ~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing data, which means it did not complete its execution to provide a result for the conjecture. | next: Re-run the test with appropriate error handling and verification of the implementation to ensure that it can handle edge cases without crashing.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17007
2	propose	ollama_remote	glm4:latest	0	0	15129
3	preregistration	ollama_remote	glm4:latest	0	0	10002
4	novelty	ollama_remote	glm4:latest	0	0	8372
5	novelty	ollama_remote	glm4:latest	0	0	8921
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14036
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9349
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6908
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8685
10	judge	ollama_remote	glm4:latest	0	0	14400

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 112808 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/027f18151318.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/027f18151318.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/027f18151318.tar.gz` (if generated)